

Overview

22 sub-chapters covering Java fundamentals through JVM internals. ShopVerse enforces strict typing, immutability via records, and concurrency primitives throughout its domain and service layers.

Collections Hierarchy

java.util.Collection (interface)

- List → ArrayList (OrderItem), LinkedList (NotificationQueue)
- Set → HashSet (ProductTag), TreeSet (CategoryService)
- Queue → ArrayBlockingQueue (AsyncNotificationQueue, cap=500)
- Deque → ArrayDeque (OrderProcessingPipeline)

java.util.Map (interface)

- LinkedHashMap – ShoppingCart (insertion-ordered)
- ConcurrentHashMap – RateLimiter (thread-safe counters)
- EnumMap – LocaleConfig (locale→bundle)

Exception Hierarchy

ShopVerseException (base, unchecked)

- DomainException
 - OrderNotFoundException
 - InsufficientInventoryException
 - ProductNotFoundException
- PaymentException
 - PaymentDeclinedException
 - PaymentTimeoutException

Java Version Features Used

Feature	Java Version	ShopVerse Usage
Records	16	All DTOs: PlaceOrderRequest, OrderResponse, ProductSummary
Sealed Classes	17	OrderEvent → OrderPlacedEvent, OrderShippedEvent, OrderCancelledEvent
Pattern Matching switch	21	EventHandler switches on sealed OrderEvent types
Text Blocks	15	Multi-line JPQL @Query annotations
Virtual Threads	21	spring.threads.virtual.enabled=true for blocking JDBC
Sequenced Collections	21	Where insertion order matters
CompletableFuture	8	NotificationService.sendAsync()
Optional<T>	8	All repository methods (no null returns in service layer)
Stream API	8	Every service method returning a list

Concurrency Primitives

Class	Where Used	Why
AtomicLong	OrderIdGenerator	Sequence without synchronization
ReentrantLock	InventoryReserveService	Replaces synchronized block
CountDownLatch	ApplicationStartupValidator	Wait for DB/Redis/Kafka ready
Semaphore	FlashSaleService	Limit concurrent purchases to 10
CopyOnWriteArrayList	EventListenerRegistry	Read-heavy listener list
BlockingQueue	AsyncNotificationQueue	Bounded at 500, feeds thread pool

JVM Flags (Dockerfile)

```
-XX:+UseG1GC -Xms512m -Xmx1g -XX:MaxGCPauseMillis=200
-XX:+HeapDumpOnOutOfMemoryError -XX:HeapDumpPath=/dumps
-XX:+UseStringDeduplication
```

Key rule: Use `char[]` (not `String`) for JWT secret in `JwtProperties.java` to avoid string pool retention.